1-3 Data Collection and Sampling Techniques

جمع البيانات وطرق المعاينة — العشوائية البسيطة والمنتظمة والطبقية والعنقودية وأخطاء المعاينة

1 · How data are collected AHMED · CLO1-C

The most common method is the **survey**, done by **telephone survey**, **mailed / emailed / online questionnaire**, or **personal interview**. Data can also come from **surveying records**, **direct observation**, **experiments**, **existing data** (online or archived), **census data**, and **simulation**.

Dr. Ahmed's sample instrument, an "Academic Performance Scale", has 8 statements rated SA / A / N / D / SD and scored 5, 4, 3, 2, 1. Each item response is **ordinal** (a Likert scale), even though it is scored with numbers.

2 · Unbiased sampling: the four methods BOTH · CLO1-C

An **unbiased sample** gives each subject in the population an equally likely chance of being selected. The four methods are **simple random**, **systematic**, **stratified** and **cluster**.

2.1 SIMPLE RANDOM SAMPLE (SRS)

Units are selected by chance, so every member of the population has a known, **equal** probability of selection. Three ways to select one: AMANI

1. **Lottery method**: number the population 1 to N, write each number on paper, mix in a bowl, draw n pieces without replacement. Disadvantages: uneven mixing causes bias; slow and impractical for large N.
2. **Random numbers by computer or calculator**: generate n different random numbers between 1 and N, without replacement.
3. **Random-number table** (Table D, Appendix A), used before computers.

USING TABLE D

(1) Number the population from 1 to N. (2) Fix the sample size n. (3) Start at any cell and move in one fixed direction (row by row or column by column). (4) Read numbers with **the same number of digits as N** (N = 950 → 3-digit groups 001–950). (5) Keep going until n different values are collected; **discard any number greater than N, and any duplicate**.

Row 2 of Table D, used by both lecturers:

```
06512 07836 88456 36313 30879 51323 76451 25578 15986 50845 57015 53684 57054 93261
```

LAPTOPS · N = 4998, N = 10 AHMED

Number the laptops 0001–4998 (4 digits) and read 4-digit groups from row 2, column 1.

DINNER INVITATION · CHOOSE 9 STUDENTS AMANI

The notes leave N blank. Solve with a class of **N = 65**: 2-digit numbers 01–65, n = 9.

Groups: 0651 2078 3688 4563 ~~6313~~ 3087 ~~9513~~ 2376 4512 ~~5578~~ 1598 ~~6508~~ 4557 0155 …

Selected:

0651, 2078, 3688, 4563, 3087, 2376, 4512

2-digit groups: 06 51 20 ~~78~~ 36 ~~88~~ 45 63 ~~63~~ 13 30 ~~87~~ ~~95~~ ~~13~~ 23 …

Selected:

06, 51, 20, 36, 45, 63, 13, 30, 23

(the second 63 and second 13 are repeats, so they are skipped). With a different N only the cut-off changes.

Table D, row 2, read in 4-digit groups (N = 4998, n = 10)

0651	**2078**	**3688**	**4563**	6313	**3087**	9513	**2376**	**4512**
				> N		> N		
5578	**1598**	6508	**4557**	**0155**	3684	5705	4932	
> N		> N			n reached	n reached	n reached	

leftover digits "61" are not a full group, so the read stops

Dr. Ahmed's laptop example. Group the digits to the number of digits in N, keep groups from 1 to N, and skip anything larger than N or repeated. The first ten accepted groups are the sample.

مثال الحواسيب: نقسم الأرقام إلى مجموعات بعدد خانات N، ونقبل ما بين 1 و N، ونتجاوز ما هو أكبر من N أو مكرر. أول عشرة مقبولة هي العينة.

2.2 SYSTEMATIC SAMPLE

Select **every k-th member** of a population arranged in some natural sequential order (a queue, a list of students, employee records), starting from a **random** point.

STEPS

$$k = \frac{N}{n} \rightarrow \text{round down to a whole number}$$

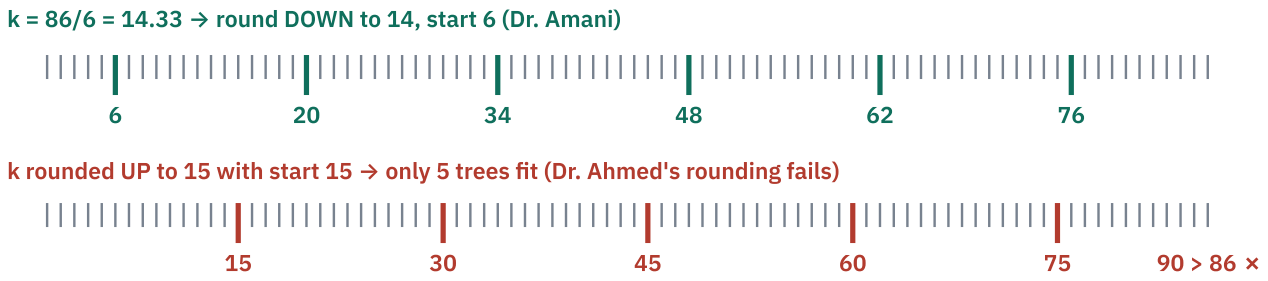
(1) Arrange the population in order. (2) Compute k. (3) Choose the first subject at random from 1 to k. (4) Keep adding k: start, start + k, start + 2k, …

86 LEMON TREES ALONG A FENCE, SELECT 6 BOTH

N = 86, n = 6, k = 86/6 = 14.33, rounded **down** to **k = 14**. Random start between 1 and 14, say 6:

6, 20, 34, 48, 62, 76

Dr. Ahmed rounds up to k = 15 with start 8: 8, 23, 38, 53, 68, 83. That happens to work, but a start of 15 gives 15, 30, 45, 60, 75 and then 90 > 86: only five trees. **Round down** and you can never fall short.

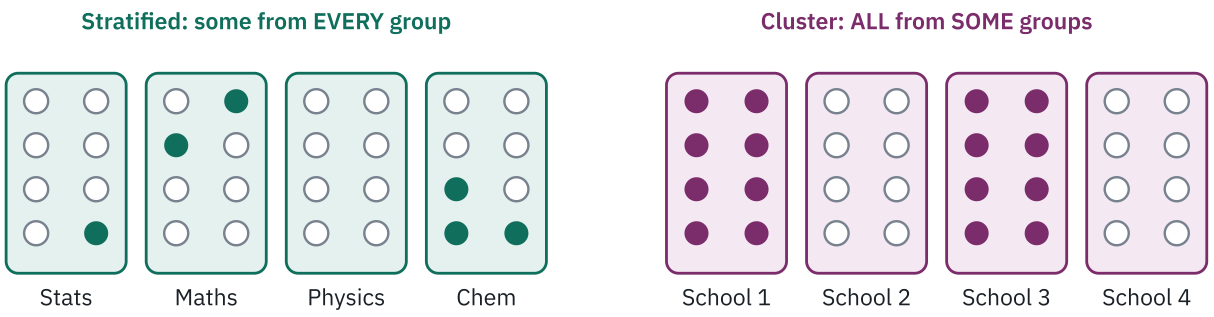


Why round k down: with k = 14, any start from 1 to 14 still reaches six trees within 86. With k = 15, a start of 15 runs past the fence after five trees, so the sample is short.

لماذا نقرّب k لأسفل؟ مع k = 14 أي بداية من 1 إلى 14 تعطي 6 أشجار ضمن 86. أما مع k = 15 فالبداية 15 تتجاوز آخر شجرة بعد 5 أشجار فقط، فتنقص العينة.

2.3 STRATIFIED AND 2.4 CLUSTER SAMPLES

	STRATIFIED طبقية	CLUSTER عنقودية
Divide into	**strata**: subgroups of individuals who share a characteristic relevant to the study	**clusters**: ready-made sections, used when the population is very large or spread over a wide area
Groups are	similar **within** each stratum	clusters similar **to each other** (homogeneous); individuals **within** a cluster diverse (heterogeneous)
Then	take an SRS from **every** stratum; sizes need not be equal (e.g. 12 from Statistics, 24 from Mathematics)	randomly choose **some** clusters and include **all** members of those clusters
Example	College of Science students, strata = departments: every department is represented	Muscat primary schools as clusters: pick 5 schools, include every student in them; or SQU colleges as clusters



Stratified: divide by a characteristic (department), then draw a random sample from every stratum, so every group is represented; sizes need not be equal. Cluster: divide into ready-made groups (schools), randomly pick whole clusters, and include everyone in them.

الطبقية: نقسم حسب خاصية (القسم) ثم نسحب عينة عشوائية من كل طبقة، فتمثَّل كل المجموعات (والأعداد لا يلزم أن تتساوى). العنقودية: نقسم إلى مجموعات جاهزة (مدارس)، نختار بعض العناقيد عشوائيًا، ونأخذ كل أفرادها.

ACTIVITY · SAMPLING METHOD AMANI

(1) In a large Muscat school, all teachers from two buildings are interviewed about homework. (2) The Oman Traffic Authority groups drivers by experience (< 5, 5–10, > 10 years) and randomly selects drivers from each group. (3) A Muscat Grand Mall supermarket surveys every 10th customer at checkout. (4) The Ministry of Health randomly generates phone numbers from a national database and calls citizens.

(1) `cluster`: buildings are the clusters, and everyone in the chosen ones is included. (2) `stratified`: a random sample from every experience group. (3) `systematic`: every k-th with k = 10. (4) `simple random`: every number in the database has an equal chance.

3 · Other (non-random) sampling methods AHMED

CONVENIENCE SAMPLE

Subjects who are convenient and readily available, easy to reach. Examples: the first 10 students entering class; interviewing people leaving a local hypermarket.

VOLUNTEER (SELF-SELECTED) SAMPLE

Respondents decide for themselves whether to be included. Examples: clinic patients who choose to fill a satisfaction survey; students who choose to complete a course evaluation.

Neither is an unbiased sample: both usually lead to a **biased** sample, because the people who are easy to reach, or who volunteer, often differ from the rest of the population.

4 · Sampling and nonsampling errors AHMED

SAMPLING ERROR

The difference between a sample result and the population result, caused only by chance in who was sampled. Example: 56% of a sample of full-time students are female, but the admissions office shows 54%. The 2% difference is sampling error.

NONSAMPLING ERROR

The data are obtained incorrectly, or the sample is biased (nonrepresentative). Examples: instrument errors (a defective thermostat reading 1.3° off), data entry or recording errors, measuring errors.

الشرح بالعربي — طرق المعاينة

(1) **العشوائية البسيطة**: كل فرد له نفس الفرصة، بالقرعة أو بالحاسوب أو بجدول الأرقام العشوائية. نقرأ مجموعات بعدد خانات N ، ونتجاوز الأكبر من N والمكرر. (2) **المنتظمة**: نقرّبه **لأسفل**، نختار بداية عشوائية من 1 إلى k ثم نضيف k كل مرة. $k = N/n$ (3) **الطبقية**: نقسم حسب خاصية، ونأخذ عينة من **كل** طبقة. (4) **العنقودية**: نقسم إلى مجموعات جاهزة، نختار **بعضها**، ونأخذ **كل** أفرادها. الفرق الذي يُسأل عنه دائمًا: الطبقية «بعض من الكل»، والعنقودية «كل من البعض». وطرق غير عشوائية: **الملاءمة** (من يسهل الوصول إليه) و**التطوعية** (من يختار المشاركة)، وكلاهما متحيز. وخطأ المعاينة = فرق ناتج عن الصدفة في اختيار العينة، أما الخطأ غير المعاينيّ فسببه خطأ في القياس أو التسجيل أو عينة متحيزة.

Chapter 1 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary*